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DECIMALS





1. Decimal Fractions

Decimal fractions are fractions where the denominator is a power of 10, like 10, 100, or 1000. Instead of writing these fractions as numbers over another number, we can use decimals.

Examples:

- $1/10$ can be written as 0.1 (zero point one).
- $3/100$ can be written as 0.03 (zero point zero three).
- $5/1000$ can be written as 0.005 (zero point zero zero five).

• Write down the decimal form of the following fractions.

$$1) \frac{7}{10} = \underline{\hspace{2cm}}$$

$$2) \frac{25}{100} = \underline{\hspace{2cm}}$$

$$3) \frac{40}{1000} = \underline{\hspace{2cm}}$$

• Convert these fractions into decimals.

$$1) \frac{2}{10} = \underline{\hspace{2cm}}$$

$$2) \frac{9}{100} = \underline{\hspace{2cm}}$$

$$3) \frac{35}{1000} = \underline{\hspace{2cm}}$$

2. Tenths, Hundredths, Thousandths

Decimals have place values, just like whole numbers. Each place after the decimal point represents a fraction of a whole.



- **Tenths (0.1):** The first digit to the right of the decimal point.
- **Hundredths (0.01):** The second digit to the right of the decimal point.
- **Thousandths (0.001):** The third digit to the right of the decimal point.
- **Write down the decimal form of the following fractions.**

$$1) \frac{6}{10} = \underline{\hspace{2cm}}$$

$$2) \frac{4}{100} = \underline{\hspace{2cm}}$$

$$3) \frac{789}{1000} = \underline{\hspace{2cm}}$$

3. Place Value Chart for Decimals

Place value helps us understand the value of each digit in a decimal number.

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths
1	2	3	.	4	5	6

- **Use a place value chart to write these numbers:**

1) 3.678

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths



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2) 45.09

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths

3) 0.304

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths

4. Adding Decimals

Decimals can be added just like whole numbers. Align the decimal points vertically before adding.

Example: $4.9 + 0.4 = 5.3$

- **Exercise:**

1) $0.6 = \frac{6}{10}$

2) $4.9 = \underline{\hspace{2cm}}$

- **Write as decimals and fractions.**

1) Three-tenths	2) Seven-tenths



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3) Five and two-tenths

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4) Two and three-tenths

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5) Eight-tenths

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6) Seven and six-tenths

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7) Nine and five-tenths

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8) Seven and one-tenth

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- Write these numbers in words.

1) 25.9 =

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2) 314.4 =

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3) 42.3 =

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4) 89.7 =

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5) 150.5 =

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- Convert the following mixed fractions to decimal form.

1) $5 \frac{4}{10} =$ _____

2) $2 \frac{7}{10} =$ _____

3) $8 \frac{1}{5} =$ _____

4) $6 \frac{3}{4} =$ _____

5) $4 \frac{9}{10} =$ _____

- **Mixed Fractions and Decimals**

A mixed fraction combines a whole number and a fraction. We can convert it to a decimal by writing the fraction as a decimal.

Example: $2 \frac{3}{10} = 2.3$

- Convert these mixed fractions to decimals

1) $5 \frac{6}{10} =$ _____

2) $3 \frac{25}{100} =$ _____



• Write these decimals as mixed fractions

$$1) \ 3.25 = \underline{\hspace{2cm}}$$

$$2) \ 9.6 = \underline{\hspace{2cm}}$$

$$3) \ 15.75 = \underline{\hspace{2cm}}$$

$$4) \ 5.4 = \underline{\hspace{2cm}}$$

$$5) \ 11.08 = \underline{\hspace{2cm}}$$

• Addition and Subtraction of Decimals (Up to 2 and 3 Decimal Places)

When adding or subtracting decimals, make sure to align the decimal points vertically before performing the calculations. This ensures each place value is correctly aligned (tenths, hundredths, thousandths).

Examples:

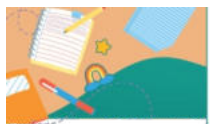
1. Addition Example:

- $5.25 + 3.75$
- Align the decimals:

• **Answer:** 9.00

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$$\begin{array}{r} 5.25 \\ + 3.75 \\ \hline 9.00 \end{array}$$



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2. Subtraction Example:

- $10.6 - 2.34$
- Align the decimals:

• **Answer:** 8.26

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$$\begin{array}{r} 10.60 \\ - 2.34 \\ \hline 8.26 \end{array}$$

• Add the following.

$$\begin{array}{r} 12.30 \\ + 7.50 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 5.80 \\ + 2.40 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 6.70 \\ + 3.20 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 3.23 \\ + 0.93 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 1.37 \\ + 0.65 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 9.99 \\ + 0.33 \\ \hline \\ \hline \end{array}$$

• Subtract the following.

$$\begin{array}{r} 7.65 \\ - 2.34 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 5.25 \\ - 0.11 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 3.02 \\ - 2.01 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 8.64 \\ - 3.34 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 6.65 \\ - 4.04 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 9.65 \\ - 5.22 \\ \hline \\ \hline \end{array}$$